Continuous Compounding & Euler's Constant e

Exponential & Logarithmic Functions • Standard: CCSS.MATH.HSF.LE.A.4

BOOK STORY OPENER: The Scottish Baron Who Doubled the Lifespan of Astronomers (John Napier), Euler's e & The Rule of 72

In 1614, John Napier at Merchiston Castle realized that logarithms turn tedious multiplication into simple height addition—the exact engineering principle of our Prime Factor blocks (where h = 15 × log2(p) mm)! Paired with Jakob Bernoulli's continuous compounding limit e and Fibonacci's Liber Abaci (1202), students model exponential doubling and half-life decay.

Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 3: Transcendentals (John Napier & Euler) & Ch. 11: Time Value of Money

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Proportional Doubling Towers & Logarithmic Blocks

Stack Fraction Blocks representing exponential doubling: 12 mm, 24 mm, 48 mm, 96 mm.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).

3. CALCULATOR ACTION (SC-32 KEYCAPS & STACK TRACE)

Keystroke Flow: [0] [•] [0] [7] [ENTER] [1] [0] [×] [e^x] [1] [0] [0] [0] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[0] [•] [0] [7]	0.07	Rate r = 0.07 entered in X
[ENTER]	0.07	Rate lifted to Y
[1] [0]	10	Time t = 10 entered in X
[×]	0.7	Exponent r × t = 0.70
[e^x]	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
[1] [0] [0] [0]	1000	Principal P = 1000 (growth factor lifts to Y)
[×]	2013.75	Total accumulated amount A = $2,013.75!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The RPN stack computes nested transcendentals from innermost exponent out to outer multiplier without parenthesis keys.

Core Investigation Problem:

A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.

Solving for Time: Doubling Time & Natural Log [LN]

Logarithms & Inverting Exponentials • Standard: CCSS.MATH.HSF.BF.B.5

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Configure stack registers on the 4-line LCD: target multiplier in Y, growth rate in X. Prediction: At 5% growth (r = 0.05), doubling time is t = ln(2) / 0.05 ~ 0.69315 / 0.05 ~ 13.86 years.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[2] [LN] [0] [•] [0] [5] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[2]	2	Doubling target factor 2
[LN]	0.6931	Natural log ln(2) = 0.693147... in X
[0] [•] [0] [5]	0.05	Rate r = 0.05 (ln(2) lifts to Y)
[÷]	13.86	t = ln(2) / r = 13.86 years!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Multiplying the base by the exponent (computing 2^5 as 2 × 5 = 10) or reversing Base and Exponent on [y^x].

Hands-On Block Remediation: Stack five 2-prime logarithmic towers to show that adding 5 tower heights multiplies 2×2×2×2×2 = 32, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

Misconception 2: Applying the power [y^x] after multiplying by principal P in $P(1 + r)^t$ (computing $(P \cdot (1+r))^t$ instead of $P \cdot ((1+r)^t)$).

Hands-On Block Remediation: Use the Inside-Out RPN rule: compute $(1 + r)^t$ first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

4. EXTENSION MATHEMATICAL INSIGHT

The natural log [LN] is the exact inverse of [e^x]. RPN preserves full floating point precision during the division.

Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Exponential growth formula: A = P × e^(r·t) for continuous growth, or A = P × (1 + r/n)^(n·t).
- Natural exponential [e^x] raises Euler's constant e to power X.
- Natural logarithm [LN] inverts exponential growth: ln(e^x) = x.

Key Sequence: `0` `•` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Problem 1: Calculate e^2 on the calculator to 4 decimal places.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 2: Calculate doubling time for an investment growing at 8% continuous interest.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 3: Compute 10^3.5 using [f:] [10^x].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 4: Evaluate log10(500) using [f:] [LOG].

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Unit 10: Exponential Growth, Compound Interest & Logarithms — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 10–12

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.

Core Solution: 40,500 bacteria. Keystrokes: [3] [ENTER] [4] [y^x] [5][0][0] [×]

Core Check Keys: 3 ENTER 4 y^x 5 0 0 ×

Extension Challenge: A radioactive isotope has decay constant k = 0.035. Find its half-life: t = ln(2) / k.

Extension Solution: 19.80 years. Keystrokes: [2] [LN] [0][.][0][3][5] [÷]

Ext. Check Keys: 2 LN 0 • 0 3 5 ÷

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Calculate e^2 on the calculator to 4 decimal places.

Expected Answer: 7.3891

SC-32 Verification Keys: 2 e^x

Problem 2: Calculate doubling time for an investment growing at 8% continuous interest.

Expected Answer: 8.66 years

SC-32 Verification Keys: 2 LN 0 • 0 8 ÷

Problem 3: Compute 10^3.5 using [f:] [10^x].

Expected Answer: 3162.28

SC-32 Verification Keys: 3 • 5 ↶ 10^x

Problem 4: Evaluate log10(500) using [f:] [LOG].

Expected Answer: 2.6990

SC-32 Verification Keys: 5 0 0 ↶ LOG

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.